1. TO FIND THE DIAMETER OF A SPHERICAL BODY

Aim:

To determine the diameter of a spherical body using Vernier Caliper.

Apparatus required:

Vernier Caliper, given spherical body (cricket ball).

Formula:

(i) Least count (LC) =1 Main scale division -1 Vernier scale division

LC = 1mm – 0.9 mm

LC = 0.1 mm (or) 0.01 cm

(ii) Diameter of the spherical object (d) = M.S.R. + (VC × LC) ± ZC cm

where, MSR - Main Scale Reading

VC - Vernier Coincide

LC - Least Count. (0.01 cm)

ZC - Zero Correction.

Procedure:

- Find the least count of the Vernier caliper.
- Find the zero correction of the Vernier caliper.
- Fix the object firmly in between the two lower jaws of the Vernier.
- Measure the main scale reading and the Vernier scale coincidence.
- Repeat the experiment by placing the jaws of the Vernier at different position of the object.
- Using the formula find the diameter of the object.

Least Count (LC): 0.01cm Zero Correction (ZC): _____

Sl. No.	Main Scale Reading (MSR) cm	Vernier coincidence (VC)	Diameter of object d = MSR + (VC × LC) ± ZC (cm)
1			
2			
3			

Average _____cm

Result: The diameter of the given spherical object (Cricket ball) is _____ cm

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